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Supplementary appendix

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Supplement to: Saad F, Clarke NW, Oya M, et al. Olaparib plus abiraterone versus placebo plus abiraterone in metastatic castration-resistant prostate cancer (PROpel): final overall survival results of a randomised, double-blind, phase 3 trial. *Lancet Oncol* 2023; published online Sept 12. [https://doi.org/10.1016/S1470-2045\(23\)00382-0](https://doi.org/10.1016/S1470-2045(23)00382-0).

Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

Supplement to Fred Saad, Noel W Clarke, Mototsugu Oya, Neal Shore, Giuseppe Procopio, João Daniel Guedes, Cagatay Arslan, Niven Mehra, Francis Parnis, Emma Brown, Friederike Schlürmann, Jae Young Joung, Mikio Sugimoto, Oliver Sartor, Yu-Zhen Liu, Christian Poehlein, Laura Barker, Paula Michelle del Rosario, Andrew J Armstrong

Olaparib plus abiraterone versus placebo plus abiraterone in metastatic castration-resistant prostate cancer (PROpel): final prespecified overall survival results of a randomised, double-blind, phase 3 trial. The Lancet Oncology 2023

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PROpel Investigators

The table below lists the principal investigator for each site who participated in the study and the number of patients randomized.

Country	Principal investigator/s	Number of patients randomised
Brazil	Brigitte M. R. H. Adam van Eyll, Leonardo Costa, Pedro Domingues, Gustavo Girotto, Juliana de Menezes, Charles Padua, Karina Vianna	100
Japan	Satoshi Anai, Takayuki Goto, Yuki Yoshi Hirayama, Taro Iguchi, Satoru Kawakami, Masashi Kato, Hidefumi Kinoshita, Takeo Kosaka, Makito Miyake, Yoshiyuki Miyazawa, Atsushi Mizokami, Masayoshi Nagata, Kazuo Nishimura, Takashi Saika, Mikio Sugimoto, Hiroyoshi Suzuki, Kazuhiro Suzuki, Kenichi Tabata, Naoki Terada, Hiroji Uemura, Kazuhiro Yoshimura	77
USA	Andrew J Armstrong, William Aronson, James Bailen, Eric Batts, Kevin Becker, Thomas Bradley, Gary Buchschacher, Kathryn Bylow, Uzair Chaudhary, William Clark, Curtis Dunshee, Chunkit Fung, Daniel Geynisman, Pooja Ghatalia, Joelle Hamilton, Clara Hwang, Lawrence Karsh, Deepak Kilari, Eli Kirshner, Eric Knoche, Michael Lilly, Luke Nordquist, Mamta Parikh, Kalpesh Patel, Christopher Pieczonka, Charles Redfern, Kristen Sanfilippo, Oliver Sartor, Nasir Shahab, Neal Shore, Justin Thomas	74
Turkey	Ahmet Alacacioglu, Catagay Arslan, Timucin Cil, Mustafa Ozguroglu, Mehmet Ali Sendur, Yuksel Urun	72
Australia	Angelyn Anton, Pretoria Bilinski, Craig Gedy, Jeffrey Goh, Anthony Joshua, Phillip Parente, Francis Parnis, Lavinia Spain, Amanda Stevanovic, Shirley Wong	62
South Korea	Young Deuk Choi, Jun Hyuk Hong, Jae Young Joung, Cheol Kwak, Tae Gyun Kwon, Ji Youl Lee	56
UK	Emma Brown, Janet Brown, Noel Clarke, Omar Din, Sophie Otter, Carla Perna, Ahmed Shaheen, John Wagstaff	49
Chile	Eugenia Lored, Mario Orio, Anibal Salazar, Eric Orellana Ulunque	44
Spain	Juan Virizuela, Daniel Castellano Gauna, Begoña Mellado González, Nuria Sala González, Maria Isabel Saez Medina, Joaquín Mateo Valderrama	43
Italy	Gaetano Facchini, Franco Nolè, Sandro Pignata, Giuseppe Procopio, Mimma Rizzo, Giorgio Vittorio Scagliotti	40
France	Abdel Rahmene Azzouzi, Pierre Bigot, Florence Joly, Denis Maillet, Vincent Massard, Friederike Schlürmann, Antoine Thiery-Vuillemin	39
Canada	Susan Ellard, Bryan Donnelly, Urban Emmenegger, Ricardo Fernandes, Geoffrey Gotto, Michael Kolinsky, Tamim Niazi, Fred Saad, Catherine Sperlich, Srikala Sridhar, Lori Wood	33
Czech Republic	Tomas Buchler, Ladislav Jarolim, Michaela Matouskova, Jiri Navratil	31
Netherlands	Niven Mehra, Robbert van Alphen, Hendrik van den Berg	28
Germany	Marinela Augustin, Susan Feyerabend, Eva Hellmis, Jörg Klier, Thomas Kretz, Stefan Machtens, Gerald Mickisch, Wolfgang Schultze-Seemann, Philipp Spiegelhalter, Friedemann Zengerling	25
Slovakia	Frederico Goncalves, Ivan Mincik, Richard Polak, Roman Sokol	18
Belgium	Nicolaas Lumen	5

Methods

Exclusion criteria: medical conditions

- Has a known additional malignancy that has had progression or has required active treatment in the last 5 years. Exceptions include basal cell carcinoma of the skin, and squamous cell carcinoma of the skin that has undergone potentially curative therapy.
- Patients with myelodysplastic syndrome (MDS)/acute myeloid leukaemia (AML) or with features suggestive of MDS/AML.
- Clinically significant cardiovascular disease as evidenced by myocardial infarction or arterial thrombotic events (eg, stroke) in the past 6 months, severe or unstable angina, atrial fibrillation or other cardiac arrhythmia requiring therapy, or New York Heart Association Class II–IV heart failure or cardiac ejection fraction measurement of <50% during screening as assessed by echocardiography or multigated acquisition scan.
- Planned or scheduled cardiac surgery or percutaneous coronary intervention procedure.
- Prior revascularisation procedure (significant coronary, carotid, or peripheral artery stenosis).
- Uncontrolled hypertension (systolic blood pressure [BP] ≥ 160 mmHg or diastolic BP ≥ 95 mmHg). Patients with a history of hypertension are allowed provided BP is controlled by antihypertensive treatment.
- History of uncontrolled pituitary or adrenal dysfunction.
- Active infection or other medical condition that would make prednisone/prednisolone use contraindicated.
- Any chronic medical condition requiring a systemic dose of corticosteroid >10 mg prednisone/prednisolone per day.
- Patients who are considered a poor medical risk due to a serious, uncontrolled medical disorder, non-malignant systemic disease or active, uncontrolled infection. Examples include, but are not limited to, uncontrolled major seizure disorder, unstable spinal cord compression, superior vena cava syndrome, active pneumonitis, extensive interstitial bilateral lung disease on high-resolution CT scan, or any psychiatric disorder that prohibits obtaining informed consent and following the study procedures.
- Persistent toxicities (Common Terminology Criteria for Adverse Events [CTCAEs] grade >2) caused by previous cancer therapy, excluding alopecia.
- Patients with brain metastases. A scan to confirm the absence of brain metastases is not required.
- Patients with spinal cord compression are excluded unless they are considered to have received definitive treatment for this and have evidence of clinically stable disease for 4 weeks.
- Patients who are unevaluable for both bone and soft tissue progression as defined by meeting both of the following criteria:
 - A bone scan referred to as a superscan showing an intense symmetric activity in the bones.
 - No soft tissue lesion (measurable or nonmeasurable) that can be assessed by RECIST.
- Patients who are unable to swallow orally administered medication and patients with gastrointestinal disorders likely to interfere with absorption of the study medication.
- Immunocompromised patients, eg, patients who are known to be serologically positive for human immunodeficiency virus.
- Patients with known active hepatitis infection (ie, hepatitis B or C).

Homologous recombination repair mutation (HRRm) status testing

Due to inherent performance characteristics of tumour tissue and plasma circulating tumour DNA (ctDNA)-based HRRm tests, and the challenges of obtaining suitable tumour tissue for next-generation sequencing, it was not possible to establish the HRRm status of all patients on PROpel using a single test modality, eg, tumour tissue testing. Tumour tissue test failure rate observed in PROpel (31·6%) was in line with that observed in PROfound (31%)¹ and other published literature.²

To ensure comprehensive evaluation of HRRm status, post hoc analysis of aggregate tumour tissue and ctDNA data was conducted to maximise the proportion of patients with assigned HRRm status and to minimise potential false negatives. Patients were classified into three groups:

- HRRm: patients with one or more HRR gene mutation detected by either test
- Non-HRRm: patients with no HRR gene mutation detected by either test, with at least one test obtaining a result
- Unknown HRRm: patients for whom mutation testing was not performed or where there was no valid result from both tests due to insufficient sample quantity or quality, or technical failure at sequencing or post-sequencing analysis steps

The aggregate tumour tissue and ctDNA data set enables a robust HRRm or BRCA mutation (BRCAm) subgroup analysis and gives confidence to the assignment of patients to the non-HRRm or non-BRCAm subgroup.

Procedure for assignment of HRR mutation status in PROpel according to FoundationOne®CDx quality control (QC) criteria and classification rules

Mutation classification was determined for HRR genes according to the Food and Drug Administration (FDA)-approved rules for the FoundationOne®CDx test as specified in the FoundationOne®CDx Summary of Safety and Effectiveness Data (FoundationOne®CDx Summaries of Safety and Effectiveness [SSED] SSED of PMA# P170019S015B). Foundation Medicine's FoundationOne®CDx test uses the COSMIC database as the basis for 'known' classifications. In addition, the Breast Cancer Information Core (BIC) database is used to define a curated list of deleterious *BRCA1/2* mutations. Details are described in the SSED of the FoundationOne®CDx Test (P170019S015B) and the FoundationFocus™ CDxBRCA Test (P160018B), respectively (FoundationFocus®CDxBRCA SSED of PMA# P160018, FoundationOne®CDx SSED of PMA# P170019S015B).

Procedure for assignment of HRR mutation status in PROpel according to FoundationOne® Liquid CDx criteria and classification rules

Mutation classification was determined according to the FDA-approved rules for the FoundationOne®Liquid CDx for *BRCA1*, *BRCA2* and *ATM* as they are specified in the SSED (FoundationOne®CDx SSED of PMA# P200016B). FoundationOne®CDx classification rules were maintained for variants identified in the other 11 HRR genes, as no prior FDA-approved process was in place for these genes (FoundationOne®CDx SSED of PMA# P170019S015B).

The biomarker rules used in PROpel are summarised in **Table S1**.

Table S1: Summary of HRR biomarker rules utilised for PROpel study

Variant class alteration type description	Alteration type	Description																														
Short variant	Nonsense, frameshift, or splice site	Any* deleterious nonsense, frameshift, or eligible splice site event occurring in the transcripts listed below: <table><tr><th>Gene symbol</th><th>RefSeq transcript</th></tr><tr><td>ATM</td><td>NM_000051</td></tr><tr><td>BRCA1</td><td>NM_007294</td></tr><tr><td>BRCA2</td><td>NM_000059</td></tr><tr><td>BARD1</td><td>NM_000465</td></tr><tr><td>BRIP1</td><td>NM_032043</td></tr><tr><td>CDK12</td><td>NM_016507</td></tr><tr><td>CHEK1</td><td>NM_001274</td></tr><tr><td>CHEK2</td><td>NM_007194</td></tr><tr><td>FANCL</td><td>NM_018062</td></tr><tr><td>PALB2</td><td>NM_024675</td></tr><tr><td>RAD51B</td><td>NM_133509</td></tr><tr><td>RAD51C</td><td>NM_058216</td></tr><tr><td>RAD51D</td><td>NM_002878</td></tr><tr><td>RAD54L</td><td>NM_003579</td></tr></table> • For <i>BRCA2</i>, truncating mutations must occur upstream of bases encoding amino acid 3326 • For <i>FANCL</i>, T367fs*13 is ineligible[†] • Eligible splice events include: – Deletion, complex indel, or base substitution events that span or occur within ±2, ±1 or 0 bases, inclusive of the intron/exon junction – Insertion events that occur adjacent to ±1 or 0 bases from the intron/exon junction – Alterations listed below	Gene symbol	RefSeq transcript	ATM	NM_000051	BRCA1	NM_007294	BRCA2	NM_000059	BARD1	NM_000465	BRIP1	NM_032043	CDK12	NM_016507	CHEK1	NM_001274	CHEK2	NM_007194	FANCL	NM_018062	PALB2	NM_024675	RAD51B	NM_133509	RAD51C	NM_058216	RAD51D	NM_002878	RAD54L	NM_003579
	Gene symbol	RefSeq transcript																														
ATM	NM_000051																															
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BRIP1	NM_032043																															
CDK12	NM_016507																															
CHEK1	NM_001274																															
CHEK2	NM_007194																															
FANCL	NM_018062																															
PALB2	NM_024675																															
RAD51B	NM_133509																															
RAD51C	NM_058216																															
RAD51D	NM_002878																															
RAD54L	NM_003579																															
	Missense or non-frameshift	List of deleterious <i>ATM</i>, <i>BRCA1</i>, and <i>BRCA2</i> mutations by protein effect <table><tr><th>ATM</th><th>BRCA1</th><th>BRCA2</th></tr><tr><td>M1T</td><td>M1V</td><td>M1R</td></tr><tr><td>R2032K</td><td>M1I</td><td>M1I</td></tr><tr><td>R2227C</td><td>C61G</td><td>V159M</td></tr></table>	ATM	BRCA1	BRCA2	M1T	M1V	M1R	R2032K	M1I	M1I	R2227C	C61G	V159M																		
ATM	BRCA1	BRCA2																														
M1T	M1V	M1R																														
R2032K	M1I	M1I																														
R2227C	C61G	V159M																														

		R2547_S2549del	C64Y	V211L
		G2765S	R71G	V211I
		R2832C	R71K	R2336P
		S2855_V2856delinsRI (annotated as S2855_V2856>RI)	R1495M	R2336H
		R3008C	E1559K	
		R3008H	D1692N	
		8418+5_8418+8de lGTGA or 8418+1_8418+4de lGTGA	D1692H	
			R1699W	
A1708E				
G1788V				
Copy number alteration	Homozygous copy number loss	- Deleterious homozygous copy number loss of one or more exons regardless of transcript - F1L CDx reports homozygous losses for only <i>BRCA1</i> and <i>BRCA2</i> - F1CDx reports homozygous losses for any HRR gene		
Rearrangement	Rearrangement	Any rearrangement that disrupts protein function regardless of transcript		

*Missense mutations in the start codon and short variant deletions spanning from upstream of the start codon annotated as M1 are biomarker negative. Non-stop extension mutations that result from insertions, deletions, or point mutations are biomarker negative. [†]FANCL T367fs*12+ germline frameshift alteration is annotated by FMI as: T367fs*13, 1099_1100insATTa (protein, CDS effect), but is considered ineligible. CDS=coding sequence. F1CDx=FoundationOne®CDx. F1L CDx=FoundationOne®Liquid CDx. FMI=Foundation Medicine Inc. HRR=homologous recombination repair.

Dose modifications for olaparib

In case a dose reduction is necessary, olaparib will be administered as described in **Table S2**.

Table S2: Dose reductions for olaparib

Dose reductions for olaparib to manage adverse events	Initial dose: 300mg twice daily Following re-challenge postinterruption (Dose reduction 1): 250 mg twice daily Dose reduction 2: 200 mg twice daily
Dose reduction for olaparib if patient develops moderate renal impairment	Initial dose: 300 mg twice daily Moderate renal impairment (calculated creatinine clearance by Cockcroft-Gault equation or based on a 24-hour urine test, between 31 and 50 mL/min) dose reduction: 200 mg twice daily
Dose reductions for olaparib if patient has to start taking a strong or moderate CYP3A inhibitor	Initial dose: 300 mg twice daily Strong CYP3A inhibitor: 100 mg twice daily Moderate CYP3A inhibitor: 150 mg twice daily

Dose modifications for abiraterone

In case dose reductions are necessary for abiraterone, the investigator should refer to abiraterone local prescribing information for further details.

Protocol deviations

Overall, 67 (8·4%) of the randomised patients had at least 1 important protocol deviation: 33 patients (8·3%) on the olaparib plus abiraterone arm and 34 patients (8·6%) on the placebo plus abiraterone arm. The most common important protocol deviation (14 patients [1·8%] overall) was ‘patient randomised but received other steroid (eg, dexamethasone) rather than prednisone/prednisolone to support abiraterone treatment prior to study treatment discontinuation’. The important protocol deviations were similar between the treatment arms, in terms of frequency and type, and they did not raise concerns with regards to study conduct. There were no important protocol deviations related to the COVID-19 pandemic.

If more than 10% of patients in either treatment arm had important protocol deviations that could affect the efficacy, this would trigger a sensitivity analysis to assess deviation bias. The proportion of patients who had at least 1 important deviation that could trigger this sensitivity analysis was 3·5% on the olaparib plus abiraterone arm and 3·3% in the placebo plus abiraterone arm. As both of these were below 10% and so did not meet the criteria to initiate the analysis, this analysis was not performed.

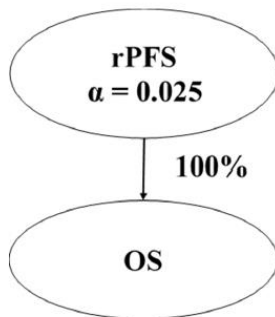
Statistical Analysis Plan

A multiple testing procedure controlled the overall one-sided type 1 error rate of 2·5%. If the primary endpoint of radiographic progression-free survival was statistically significant, then overall survival (OS) would be tested in a hierarchical fashion (**figure S1**).

The one-sided boundary for statistical significance of rPFS at interim analysis based upon the observed 394 events was 0·0162.

The O’Brien and Fleming spending function,³ calculated based upon actual observed events, was used to control the overall type 1 error, with the restriction that alpha spend for the OS interim analysis at the first data cut-off would not exceed 0·0005.

Figure S1: Multiplicity strategy maintaining overall type 1 error rate.



OS=overall survival. rPFS=radiographic progression-free survival.

Prespecified subgroups

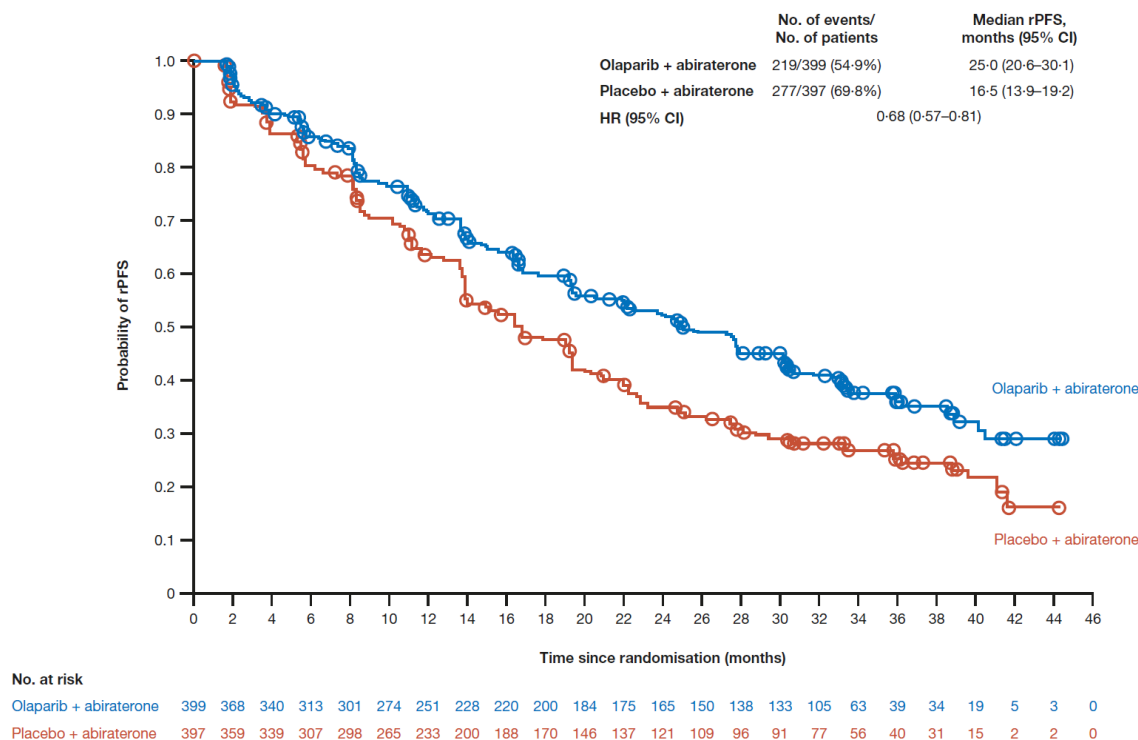
Analysis of rPFS and OS was prespecified for the following subgroups:

- Metastasis (bone only, visceral, or other)
- Docetaxel treatment at metastatic hormone-sensitive prostate cancer (mHSPC) stage (yes or no)
- HRRm status subgroup (HRRm, non-HRRm, unknown) based on a ctDNA-based test (FoundationOne®Liquid CDx)
- HRRm status subgroup (HRRm, non-HRRm, unknown) based on a tissue test (FoundationOne®CDX)
- HRRm status subgroup (HRRm, non-HRRm, unknown) based on a germline blood test (Myriad MyRisk™) (when data are available)
- Eastern Cooperative Oncology Group (ECOG) performance status at baseline (0 or 1)
- Age at randomization (<65 years, ≥65 years)
- Region (Asia, Europe, North and South America)
- Race (White, Black/African American, Asian, Other)
- Baseline prostate-specific antigen (PSA) (above/below median baseline PSA of the patients across both treatment groups)

Results

Figure S2: Kaplan–Meier estimates of updated radiographic progression-free survival by investigator assessment in the ITT population at final prespecified OS analysis (DCO: 12 October 2022).

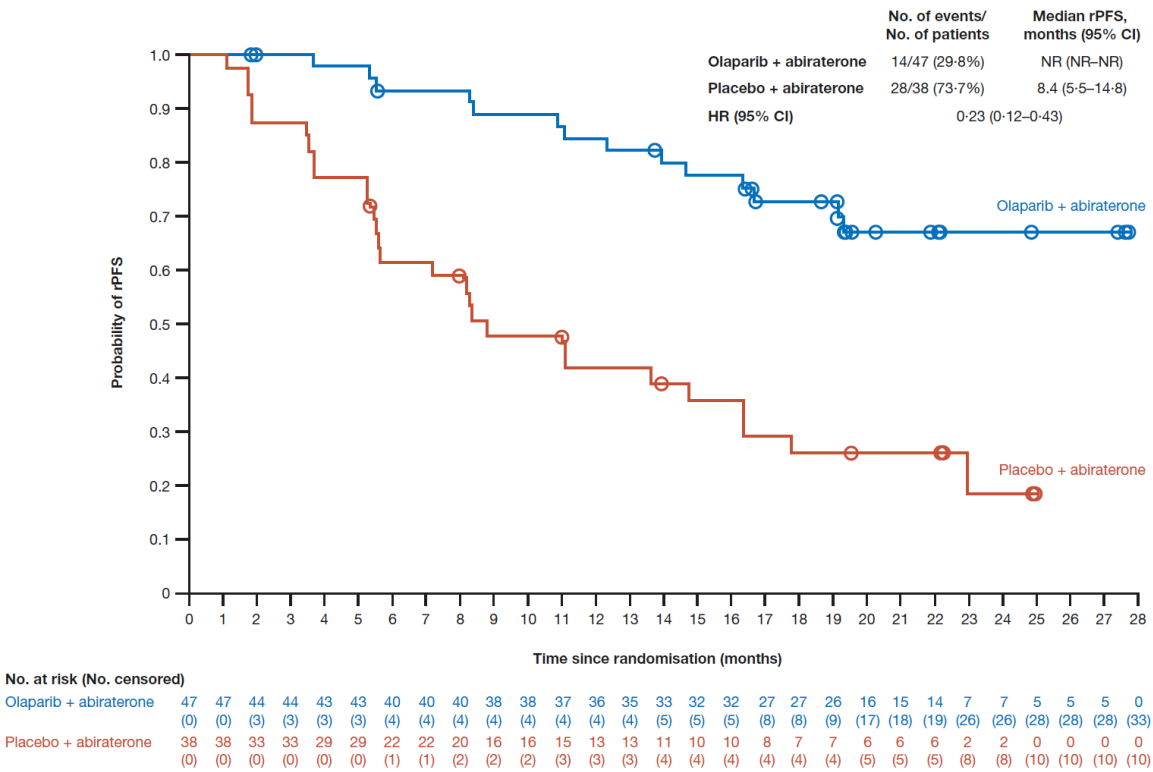
rPFS analysis at this final prespecified OS analysis was descriptive as rPFS at the primary analysis was statistically significant.



DCO October 12, 2022. There were 219 progression or death events in the olaparib plus abiraterone arm and 277 in the placebo plus abiraterone arm. The median (range) duration of follow-up for disease progression in patients with censored data was 32.5 (0.03–44.32) months in the olaparib plus abiraterone arm and 33.0 (0.03–44.25) months in the placebo plus abiraterone arm. A circle indicates a censored observation. CI=confidence interval. DCO=data cutoff. HR=hazard ratio. ITT=intention-to-treat. OS=overall survival. rPFS=radiographic progression-free survival.

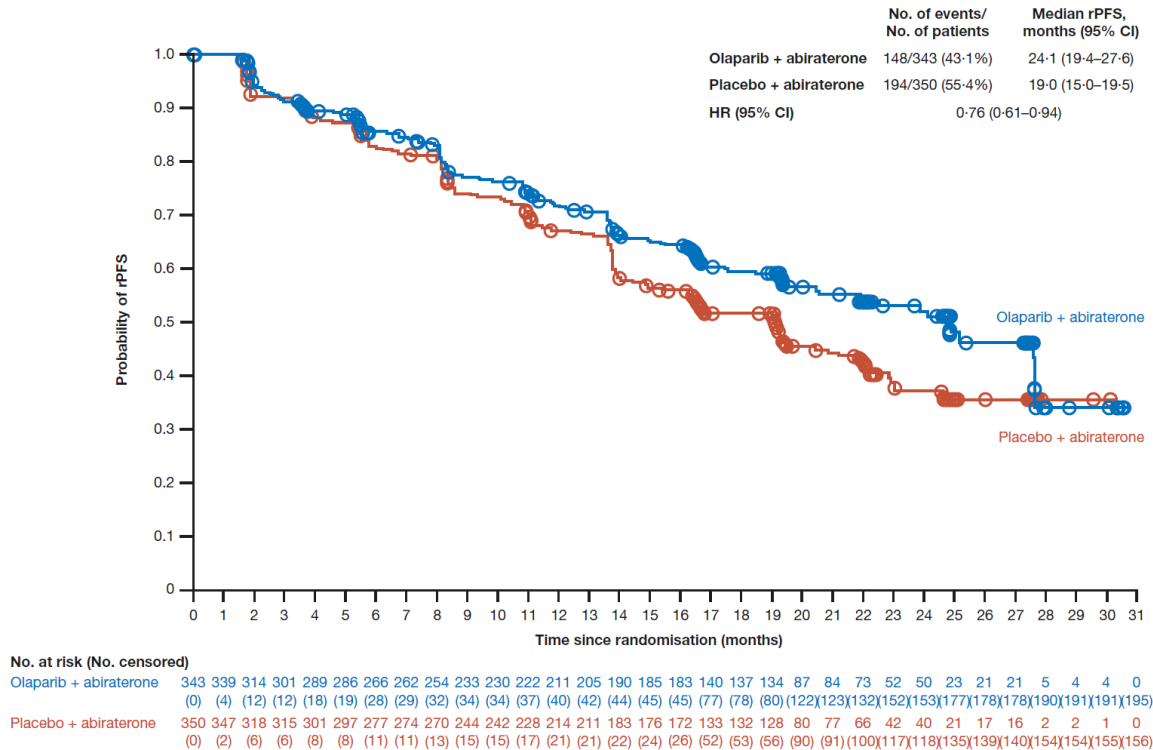
Figure S3: Kaplan–Meier estimates of radiographic progression-free survival in the BRCAm and non-BRCAm population (DCO: 30 July 2021).

A. Radiographic progression-free survival in the BRCAm subgroup



Data cutoff 30 July 2021. There were 14 progression or death events in the olaparib plus abiraterone arm and 28 in the placebo plus abiraterone arm. A circle indicates a censored observation.

B. Radiographic progression-free survival in the non-BRCAm subgroup



Data cutoff 30 July 2021. There were 148 progression or death events in the olaparib plus abiraterone arm and 194 in the placebo plus abiraterone arm. A circle indicates a censored observation.

Table S3: Therapies post-discontinuation of study treatment (DCO: 12 October 2022).

	Olaparib plus abiraterone (N=399)	Placebo plus abiraterone (N=397)
Any post-discontinuation anti-cancer therapy	179 (44.9)	216 (54.4)
Cytotoxic chemotherapy	123 (30.8)	167 (42.1)
Docetaxel	95 (23.8)	141 (35.5)
Cabazitaxel	43 (10.8)	62 (15.6)
Carboplatin	9 (2.3)	10 (2.5)
Cisplatin	2 (0.5)	4 (1.0)
Other	5 (1.3)	13 (3.3)
Hormonal therapy	67 (16.8)	75 (18.9)
Abiraterone	23 (5.8)	19 (4.8)
Abiraterone acetate	5 (1.3)	5 (1.3)
Enzalutamide	39 (9.8)	48 (12.1)
Other	6 (1.5)	13 (3.3)
Immunotherapy	23 (5.8)	23 (5.8)
Targeted therapy	20 (5.0)	29 (7.3)
Radium 223 dichloride	10 (2.5)	15 (3.8)
Lutetium (Lu 177)	6 (1.5)	1 (0.3)
Other	5 (1.3)	15 (3.8)
PARP inhibitor	2 (0.5)	5 (1.3)
Other	8 (2.0)	19 (4.8)

Patients could be counted in more than one anti-cancer therapy. DCO=data cutoff. PARP=poly (ADP-ribose) polymerase.

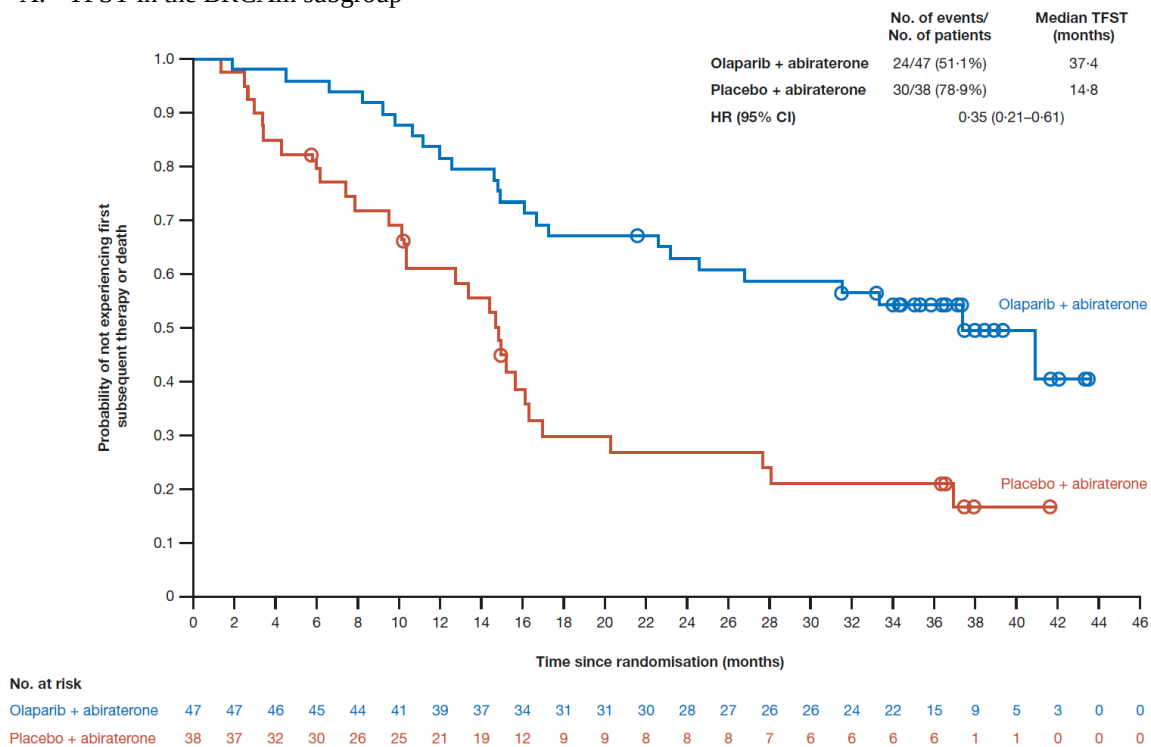
Table S4: Circulating tumour cell counts and conversion rates (DCO: 30 July 2021).

		Olaparib plus abiraterone (N=399)	Placebo plus abiraterone (N=397)
Number of patients with CTCs from ≥5 cells/7.5 mL at baseline	n	103	97
Number of patients with CTC conversion*	n	58	44
	Proportion of patients (95% CI) [†]	56.3 (46.2–66.1)	45.4 (35.2–55.8)

*CTC conversion defined as the proportion of patients who achieved a decline in CTCs from ≥5 cells/7.5 mL blood at baseline to <5 cells/7.5 mL at any time post-baseline, which is prior to subsequent anti-cancer therapy. [†]Denominator includes the subset of the patients who had CTCs ≥5 cells/7.5 mL at baseline. CI calculated using Clopper-Pearson exact method for binomial proportion. CI=confidence interval. CTC=circulating tumour cell.

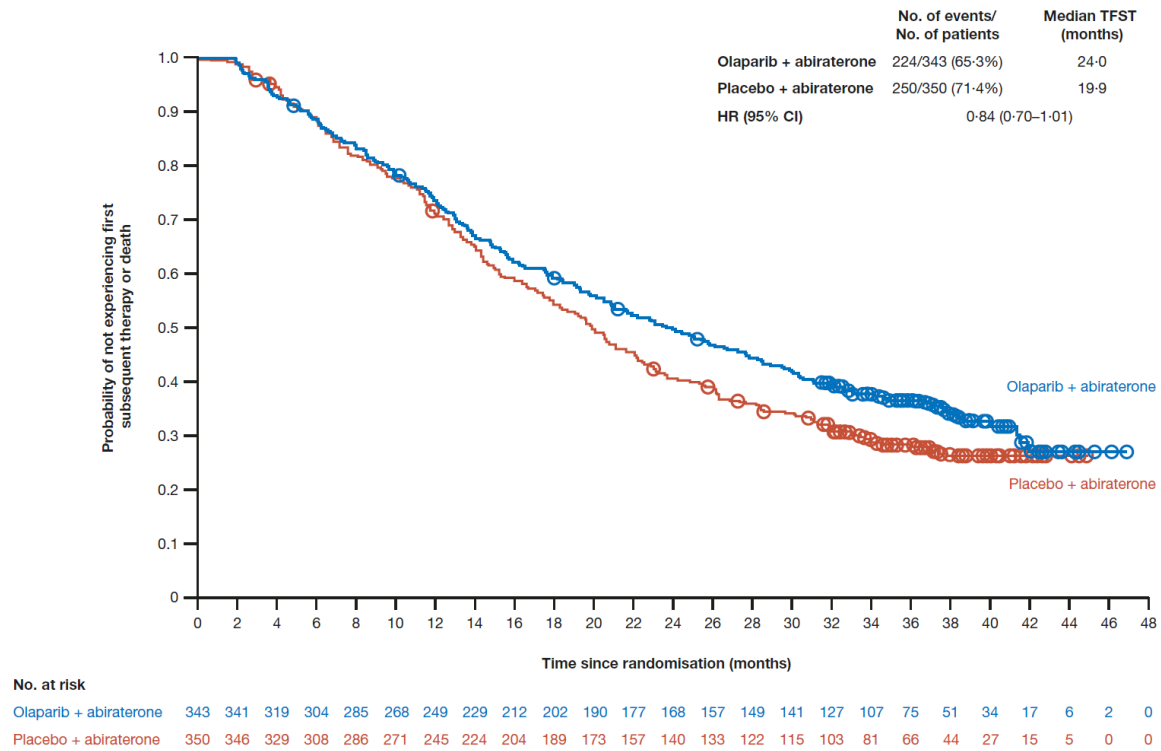
Figure S4: Kaplan–Meier estimates of TFST, PFS2 and time to PSA progression in BRCAm and non-BRCAm subgroups (DCO: 12 October 2022).

A. TFST in the BRCAm subgroup



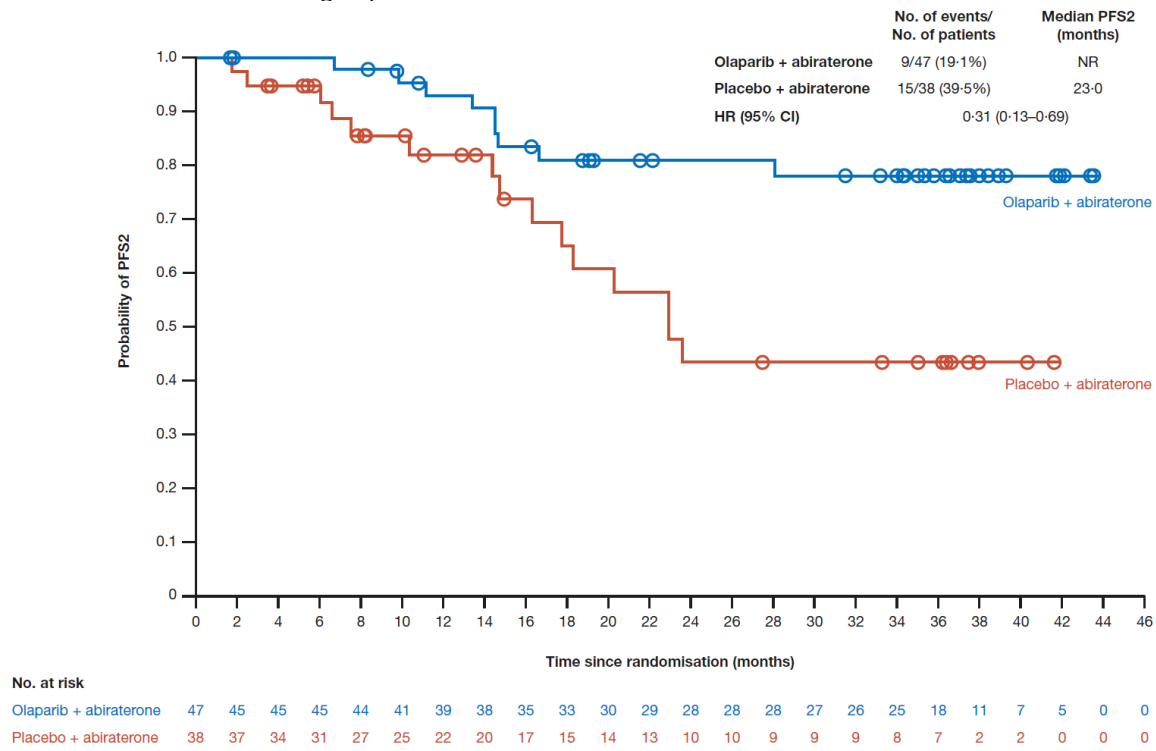
DCO October 12, 2022. There were 24 patients who received a first subsequent therapy in the olaparib plus abiraterone arm, and 30 patients in the placebo plus abiraterone arm. Time to first subsequent therapy (excluding radiotherapy) was defined as the time from randomisation to the start date of the first subsequent anti-cancer therapy after discontinuation of randomised treatment or death from any cause (whichever was earlier). Any patient not known to have died at the time of the analysis and not known to have had a subsequent therapy was censored at the last known time to have not received first subsequent therapy. BRCAm=BRCA mutation. CI=confidence interval. DCO=data cutoff. HR=hazard ratio. PFS2=time to second progression. PSA=prostate-specific antigen. TFST=time to first subsequent therapy.

B. TFST in the non-BRCAm subgroup



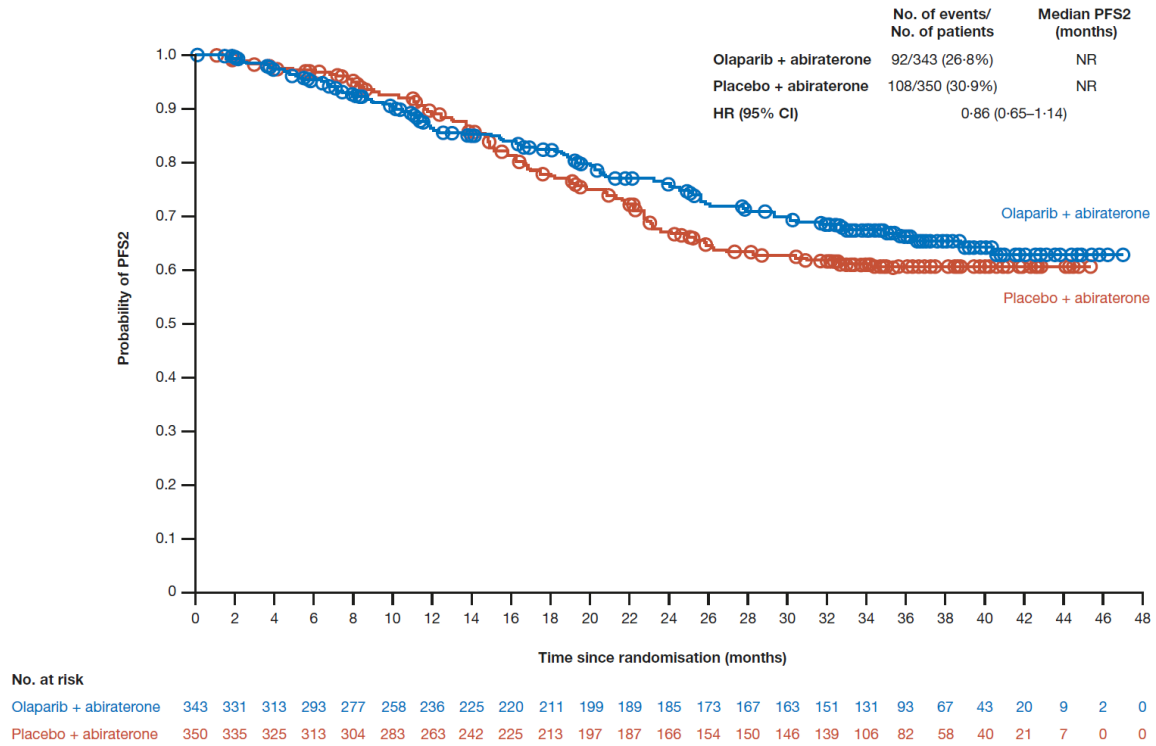
Data cutoff 12 October 2022. There were 224 patients who received a first subsequent therapy in the olaparib plus abiraterone arm, and 250 patients in the placebo plus abiraterone arm. Time to first subsequent therapy (excluding radiotherapy) was defined as the time from randomisation to the start date of the first subsequent anti-cancer therapy after discontinuation of randomised treatment or death from any cause (whichever was earlier). Any patient not known to have died at the time of the analysis and not known to have had a subsequent therapy was censored at the last known time to have not received first subsequent therapy. BRCAm=BRCA mutation. CI=confidence interval. HR=hazard ratio. TFST=time to first subsequent therapy.

C. PFS2 in the BRCAm subgroup



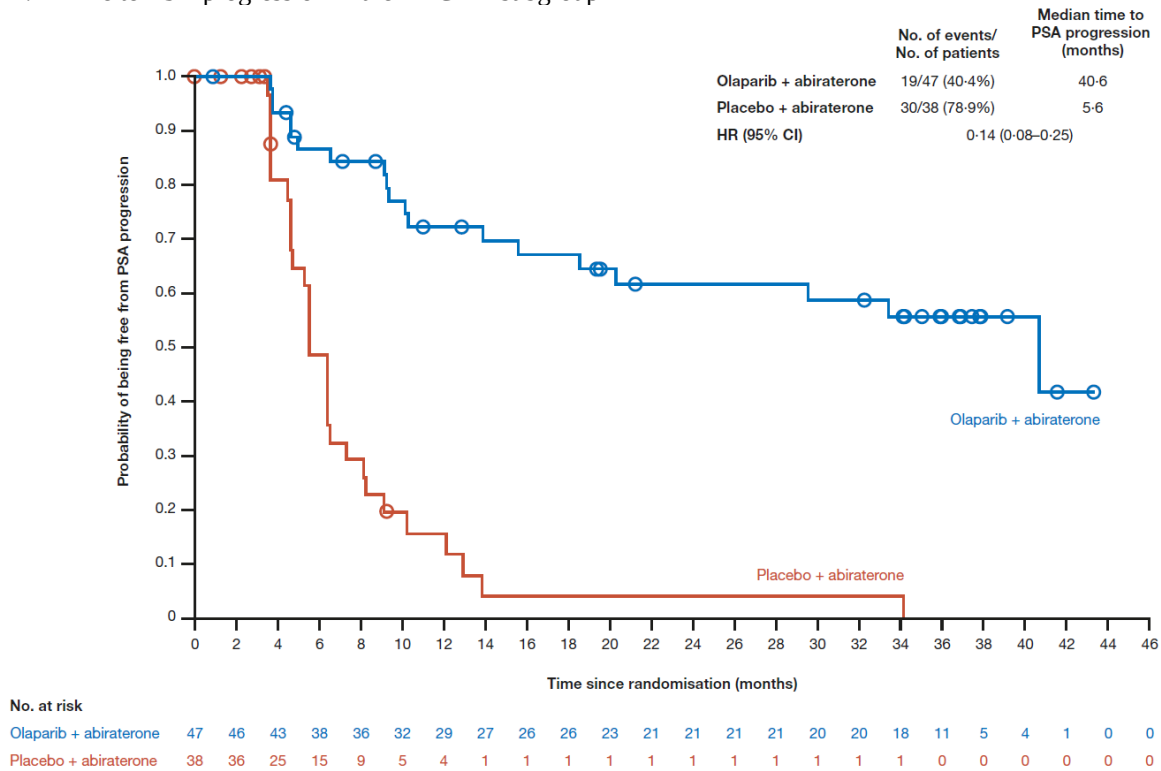
Data cutoff 12 October 2022. There were nine second progression or death events in the olaparib plus abiraterone arm and 15 in the placebo plus abiraterone arm. Patients who had not had a second disease progression event or died at the time of analysis, or who had second progression or died after two or more missed visits, were censored at the latest evaluable assessment when they were known to be alive and without a second disease progression. BRCAm=BRCA mutation. CI=confidence interval. HR=hazard ratio. NR=not reached. PFS2=time to second progression.

D. PFS2 in the non-BRCAm subgroup



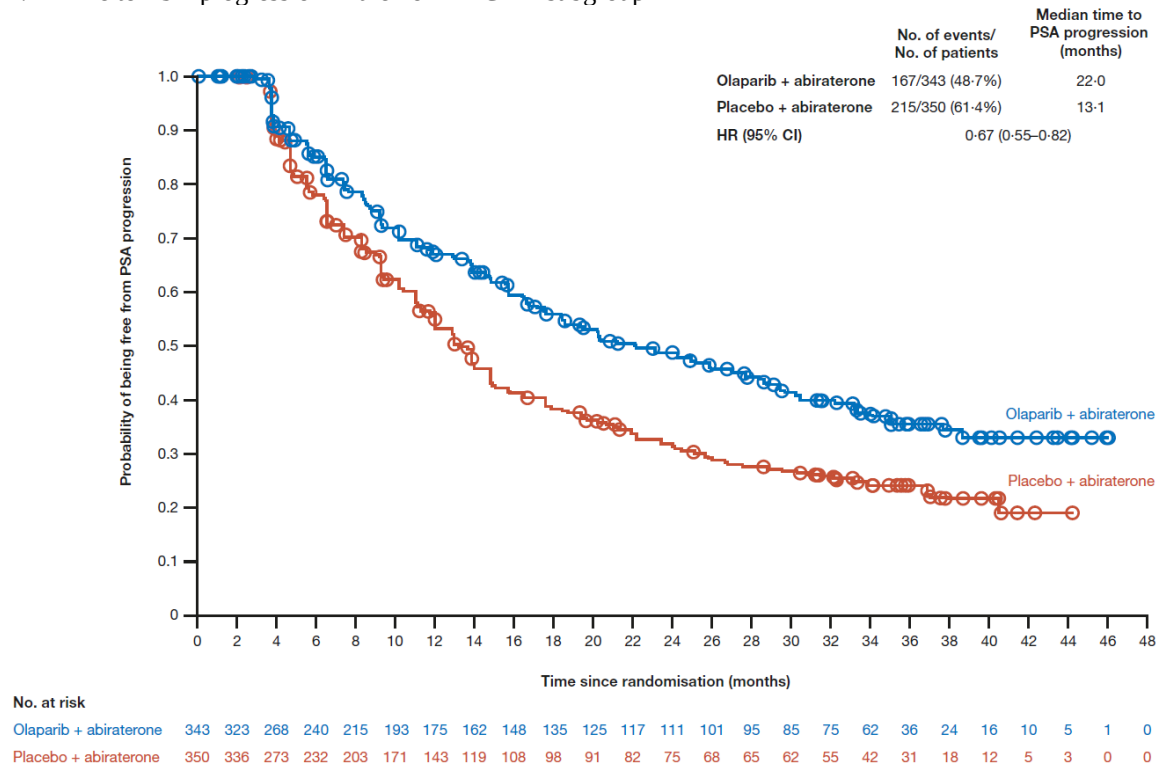
Data cutoff 12 October 2022. There were 92 second progression or death events in the olaparib plus abiraterone arm and 108 in the placebo plus abiraterone arm. Patients who had not had a second disease progression event or died at the time of analysis, or who had second progression or died after two or more missed visits, were censored at the latest evaluable assessment when they were known to be alive and without a second disease progression. BRCAm=BRCA mutation. CI=confidence interval. HR=hazard ratio. NR=not reached. PFS2=time to second progression.

E. Time to PSA progression in the BRCAm subgroup



Data cutoff 12 October 2022. There were 19 PSA progression or death events in the olaparib plus abiraterone arm and 30 in the placebo plus abiraterone arm. Time to PSA progression was defined as the time from randomisation to first PSA progression per the Prostate Cancer Working Group 3 criteria. BRCAm=BRCA mutation. CI=confidence interval. HR=hazard ratio. PSA=prostate-specific antigen.

F. Time to PSA progression in the non-BRCAm subgroup



Data cutoff 12 October 2022. There were 167 PSA progression or death events in the olaparib plus abiraterone arm and 215 in the placebo plus abiraterone arm. Time to PSA progression was defined as the time from randomisation to first PSA progression per the Prostate Cancer Working Group 3 criteria. BRCAm=BRCA mutation. CI=confidence interval. HR=hazard ratio. PSA=prostate-specific antigen.

Table S5: All deaths (DCO: 12 October 2022).

	Olaparib plus abiraterone (N=398)	Placebo plus abiraterone (N=396)
Total number of deaths, n (%)	176 (44.1)	205 (51.6)
Related to disease under investigation only	127 (31.8)	162 (40.8)
AE with outcome of death only	23 (5.8)	13 (3.3)
AE with outcome of death only (AE start date after 30-day follow-up period)	1 (0.3)	0
Related to disease under investigation and an AE with outcome of death	3 (0.8)	7 (1.8)
Other deaths*	22 (5.5)	23 (5.8)

*Patients who died and are not captured in the earlier categories. Death related to disease under investigation is determined by the investigator. Rows are mutually exclusive; patients are only reported in 1 category. DCO=data cutoff.

Table S6: Adverse events of cardiac failure (DCO: 12 October 2022).

	Olaparib plus abiraterone (N=398)	Placebo plus abiraterone (N=396)
Cardiac failure events, n (%)	7 (1.8)	7 (1.8)
Acute pulmonary oedema	0	2 (0.5)
Cardiac failure	3 (0.8)	2 (0.5)
Cardiac failure acute	1 (0.3)	0
Cardiac failure chronic	0	1 (0.3)
Cardiac failure congestive	1 (0.3)	2 (0.5)
Ejection fraction decreased	1 (0.3)	0
Left ventricular failure	2 (0.5)	0
Pulmonary oedema	0	2 (0.5)

Patients were counted once for each type of adverse event. DCO=data cutoff.

Table S7: Embolic and thrombotic adverse events (DCO: 12 October 2022).

	Olaparib plus abiraterone (N=398)	Placebo plus abiraterone (N=396)
Embolic and thrombotic events, arterial n (%)	10 (2.5)	14 (3.5)
Acute coronary syndrome	0	1 (0.3)
Acute myocardial infarction	4 (1.0)	3 (0.8)
Ischaemic stroke	1 (0.3)	4 (1.0)
Myocardial infarction	1 (0.3)	4 (1.0)
Peripheral arterial occlusive disease	3 (0.8)	0
Peripheral embolism	0	1 (0.3)
Transient ischaemic attack	1 (0.3)	1 (0.3)
Embolic and thrombotic events, venous n (%)	34 (8.5)	16 (4.0)
Deep vein thrombosis	10 (2.5)	3 (0.8)
Portal vein thrombosis	0	1 (0.3)
Pulmonary embolism	29 (7.3)	9 (2.3)
Retinal vein occlusion	0	1 (0.3)
Superficial vein thrombosis	0	2 (0.5)

Patients were counted once for each type of adverse event. DCO=data cutoff.

Table S8: COVID-19-related adverse events (DCO: 12 October 2022).

Of patients who developed COVID-19, 44 had a COVID-19 vaccine administered before diagnosis with COVID-19. Within the olaparib plus abiraterone arm, 13 patients were from Brazil, eight were from Turkey, and 11 were from the United States.

	Olaparib plus abiraterone (N=398)	Placebo plus abiraterone (N=396)
Any AE, n (%)	63 (15·8)	39 (9·8)
Any AE grade ≥3	23 (5·8)	11 (2·8)
Any AE with outcome of death	12 (3·0)	3 (0·8)
Any SAE (including events with outcome of death)	24 (6·0)	13 (3·3)
Any AE leading to discontinuation of olaparib/placebo	3 (0·8)	0
Any AE leading to dose reduction of olaparib/placebo	0	0
Any AE leading to interruption of olaparib/placebo	28 (7·0)	11 (2·8)

AE=adverse event. COVID-19=coronavirus disease 2019. DCO=data cutoff. SAE=serious adverse event.

Table S9: Myelodysplastic syndrome (DCO: 12 October 2022).

Case 1
During hospitalisation for COVID-19, the patient developed pancytopenia, which was considered by the investigator to be CTCAE Grade 4 MDS. Symptoms included weakness, headache, light headedness, shortness of breath, paleness, fever, and fatigue. Although a bone marrow biopsy was performed, the material contained only clots and could not be well examined. No treatment was reported for the event of MDS. Action taken with olaparib and abiraterone in response to the event of MDS was reported as none as the patient had previously discontinued study treatment due to the event of COVID-19 pneumonia. The investigator considered the event of MDS to be related to olaparib and unrelated to abiraterone. The patient died prior to DCO2. The primary cause of death was COVID-19 pneumonia. While pancytopenia is a possible sign of MDS, it is also a known complication of COVID-19, and the lack of bone marrow confirmation of the diagnosis in this case precludes a meaningful assessment of the contribution of olaparib in the development of these events.
Case 2
The patient was diagnosed with CTCAE Grade 4 SAE of MDS. Olaparib was withdrawn in response to the event of MDS. The investigator considered the event of MDS to be related to olaparib and unrelated to abiraterone. The patient was alive at DCO3 and the event of MDS was reported as not recovered.

COVID-19=coronavirus disease 2019. CTCAE=Common Terminology Criteria for Adverse Events. DCO=data cut-off. MDS=myelodysplastic syndrome. SAE=serious adverse event.

Table S10: New primary malignancies (DCO: 12 October 2022).

	Olaparib plus abiraterone (N=398)	Placebo plus abiraterone (N=396)
Any new primary malignancy, n (%)	18 (4·5)	14 (3·5)
Bladder cancer	3 (0·8)	0
Colon cancer	3 (0·8)	1 (0·3)
Colorectal cancer	0	1 (0·3)
Adenocarcinoma of colon	0	1 (0·3)
Diffuse large B-cell lymphoma	0	1 (0·3)
Gastric cancer	1 (0·3)	2 (0·5)
Lung adenocarcinoma	1 (0·3)	2 (0·5)
Malignant melanoma	1 (0·3)	0
Malignant melanoma <i>in situ</i>	1 (0·3)	0
Neoplasm skin	1 (0·3)	0
Neuroendocrine carcinoma of skin	1 (0·3)	0
Neuroendocrine carcinoma	0	1 (0·3)
Non-small cell lung cancer	0	2 (0·5)
Oesophageal neoplasm	0	1 (0·3)
Oropharyngeal cancer	0	1 (0·3)
Plasma cell myeloma	1 (0·3)	0
Rectal cancer	2 (0·5)	0
Small intestine adenocarcinoma	1 (0·3)	0
Squamous cell carcinoma of the tongue	1 (0·3)	0
Transitional cell carcinoma	1 (0·3)	1 (0·3)

DCO=data cutoff.

Table S11: All grade 3, 4, and 5 adverse events (safety analysis set; DCO: 12 October 2022)

Adverse event	Olaparib plus abiraterone (n=398)					Placebo plus abiraterone (n=396)				
	All grade	Grade 1–2	Grade 3	Grade 4	Grade 5	All grade	Grade 1–2	Grade 3	Grade 4	Grade 5
Any adverse event	389 (97·7)	167 (42·0)	164 (41·2)	32 (8·0)	26 (6·5)	380 (96·0)	209 (52·8)	135 (34·1)	16 (4·0)	20 (5·1)
Anaemia [†]	198 (49·7)	159 (39·9)	61 (15·3)	4 (1·0)	..	70 (17·7)	57 (14·4)	13 (3·3)	..	..
Fatigue or asthenia	154 (38·7)	150 (37·7)	10 (2·5)	..	..	120 (30·3)	117 (29·5)	6 (1·5)	..	..
Nausea	122 (30·7)	121 (30·4)	1 (0·3)	..	..	57 (14·4)	56 (14·1)	1 (0·3)	..	..
Back pain	86 (21·6)	82 (20·6)	4 (1·0)	..	..	79 (19·9)	73 (18·4)	6 (1·5)	..	..
Diarrhoea	82 (20·6)	77 (19·3)	5 (1·3)	..	..	42 (10·6)	41 (10·4)	1 (0·3)	..	..
Constipation	74 (18·6)	74 (18·6)	..	..	..	59 (14·9)	58 (14·6)	1 (0·3)	..	..
Decreased appetite	66 (16·6)	62 (15·6)	4 (1·0)	..	..	31 (7·8)	31 (7·8)	..	..	..
Vomiting	62 (15·6)	56 (14·1)	6 (1·5)	..	..	37 (9·3)	36 (9·1)	1 (0·3)	..	..
Hypertension	61 (15·3)	46 (11·6)	15 (3·8)	..	..	74 (18·7)	56 (14·1)	18 (4·5)	..	..
Arthralgia	58 (14·6)	58 (14·6)	..	..	..	77 (19·4)	75 (18·9)	2 (0·5)	..	..
COVID-19	51 (12·8)	36 (9·0)	7 (1·8)	1 (0·3)	7 (1·8)	35 (8·8)	27 (6·8)	4 (1·0)	1 (0·3)	3 (0·8)
Peripheral oedema	49 (12·3)	49 (12·3)	..	..	..	50 (12·6)	49 (12·4)	1 (0·3)	..	..
Dizziness	49 (12·3)	49 (12·3)	..	..	..	27 (6·8)	27 (6·8)	..	..	..
Urinary tract infection	46 (11·6)	36 (9·0)	10 (2·5)	..	..	35 (8·8)	31 (7·8)	4 (1·0)	..	..
Cough	47 (11·8)	47 (11·8)	..	..	..	29 (7·3)	29 (7·3)	..	..	..
Urinary tract infection	46 (11·6)	36 (9·0)	10 (2·5)	..	..	35 (8·8)	31 (7·8)	4 (1·0)	..	..
Neutropenia [‡]	40 (10·1)	24 (6·0)	14 (3·5)	6 (1·5)	..	14 (3·5)	7 (1·8)	7 (1·8)	..	..
Headache	39 (9·8)	38 (9·5)	1 (0·3)	..	..	26 (6·6)	26 (6·6)	..	..	..
Dyspnoea	39 (9·8)	37 (9·3)	2 (0·5)	..	..	27 (6·8)	27 (6·8)	..	..	..
Pain in extremity	38 (9·5)	37 (9·3)	1 (0·3)	..	..	38 (9·6)	37 (9·3)	1 (0·3)	..	..
Hot flush	35 (8·8)	35 (8·8)	..	..	..	51 (12·9)	51 (12·9)	..	..	..
Leukopenia [§]	35 (8·8)	27 (6·8)	8 (2·0)	5 (1·3)	..	12 (3·0)	11 (2·8)	2 (0·5)	..	..
Thrombocytopenia [§]	27 (6·8)	25 (6·3)	2 (0·5)	1 (0·3)	..	17 (4·3)	15 (3·8)	1 (0·3)	1 (0·3)	..
Hypokalaemia	34 (8·5)	27 (6·8)	4 (1·0)	3 (0·8)	..	18 (4·5)	15 (3·8)	3 (0·8)	..	..
Lymphocyte count decreased	34 (8·5)	19 (4·8)	12 (3·0)	3 (0·8)	..	17 (4·3)	11 (2·8)	6 (1·5)	..	..
Insomnia	31 (7·8)	30 (7·5)	1 (0·3)	..	..	28 (7·1)	28 (7·1)	..	..	..
Hyperglycaemia	30 (7·5)	22 (5·5)	8 (2·0)	..	..	28 (7·1)	22 (5·6)	5 (1·3)	1 (0·3)	..
Pyrexia	29 (7·3)	28 (7·0)	1 (0·3)	..	..	21 (5·3)	21 (5·3)	..	..	..
Fall	29 (7·3)	28 (7·0)	1 (0·3)	..	..	29 (7·3)	28 (7·0)	1 (0·3)	..	..
Pulmonary embolism	29 (7·3)	..	26 (6·5)	2 (0·5)	1 (0·3)	9 (2·3)	..	7 (1·8)	2 (0·5)	..
Blood creatinine increased	28 (7·0)	26 (6·5)	2 (0·5)	..	..	19 (4·8)	18 (4·5)	1 (0·3)	..	..
Abdominal pain	27 (6·8)	27 (6·8)	..	..	..	16 (4·0)	14 (3·5)	2 (0·5)	..	..
Weight decreased	27 (6·8)	25 (6·3)	2 (0·5)	..	..	15 (3·8)	15 (3·8)	..	..	..
Pneumonia	25 (6·3)	15 (3·8)	6 (1·5)	3 (0·8)	1 (0·3)	13 (3·3)	9 (2·3)	4 (1·0)	..	..
Hypertriglyceridaemia	24 (6·0)	19 (4·8)	5 (1·3)	..	..	18 (4·5)	16 (4·0)	2 (0·5)	..	..
Lymphopenia	24 (6·0)	18 (4·5)	6 (1·5)	..	..	10 (2·5)	6 (1·5)	4 (1·0)	..	..
Atrial fibrillation	22 (5·5)	16 (4·0)	6 (1·5)	..	..	12 (3·0)	9 (2·3)	3 (0·8)	..	..
Blood alkaline phosphatase increased	22 (5·5)	19 (4·8)	1 (0·3)	2 (0·5)	..	21 (5·3)	15 (3·8)	6 (1·5)	..	..
Haematuria	19 (4·8)	18 (4·5)	1 (0·3)	..	..	14 (3·5)	13 (3·3)	1 (0·3)	..	..
Hypocalcaemia	18 (4·5)	16 (4·0)	1 (0·3)	1 (0·3)	..	14 (3·5)	11 (2·8)	2 (0·5)	1 (0·3)	..
Blood bilirubin increased	18 (4·5)	17 (4·3)	1 (0·3)	..	..	9 (2·3)	8 (2·0)	1 (0·3)	..	..
Bone pain	17 (4·3)	17 (4·3)	..	..	..	7 (1·8)	5 (1·3)	2 (0·5)	..	..

Aspartate aminotransferase increased	16 (4·0)	14 (3·5)	2 (0·5)	..	..	23 (5·8)	19 (4·8)	4 (1·0)	..	..
Neck pain	16 (4·0)	14 (3·5)	2 (0·5)	..	..	7 (1·8)	7 (1·8)	..	..	..
Alanine aminotransferase increased	15 (3·8)	11 (2·8)	4 (1·0)	..	..	28 (7·1)	19 (4·8)	8 (2·0)	1 (0·3)	..
Amylase increased	15 (3·8)	11 (2·8)	3 (0·8)	1 (0·3)	..	19 (4·8)	14 (3·5)	5 (1·3)	..	..
Malaise	15 (3·8)	14 (3·5)	1 (0·3)	..	..	10 (2·5)	10 (2·5)	..	..	..
Electrocardiogram QT prolonged	14 (3·5)	11 (2·8)	3 (0·8)	..	..	2 (0·5)	1 (0·3)	1 (0·3)	..	..
Rash	13 (3·3)	13 (3·3)	..	..	..	15 (3·8)	14 (3·5)	1 (0·3)	..	..
Acute kidney injury	13 (3·3)	8 (2·0)	4 (1·0)	1 (0·3)	..	9 (2·3)	5 (1·3)	3 (0·8)	..	1 (0·3)
Muscular weakness	13 (3·3)	12 (3·0)	1 (0·3)	..	..	7 (1·8)	7 (1·8)	..	..	..
Anxiety	12 (3·0)	11 (2·8)	1 (0·3)	..	..	12 (3·0)	12 (3·0)	..	..	..
Dehydration	12 (3·0)	10 (2·5)	2 (0·5)	..	..	3 (0·8)	2 (0·5)	1 (0·3)	..	..
Depression	11 (2·8)	11 (2·8)	..	..	..	11 (2·8)	10 (2·5)	1 (0·3)	..	..
Hypophosphataemia	11 (2·8)	9 (2·3)	1 (0·3)	1 (0·3)	..	8 (2·0)	5 (1·3)	3 (0·8)	..	..
Musculoskeletal pain	11 (2·8)	10 (2·5)	1 (0·3)	..	..	6 (1·5)	5 (1·3)	1 (0·3)	..	..
Vertigo	11 (2·8)	10 (2·5)	1 (0·3)	..	..	5 (1·3)	5 (1·3)	..	..	..
Gastroenteritis	11 (2·8)	10 (2·5)	1 (0·3)	..	..	3 (0·8)	3 (0·8)	..	..	..
Urinary incontinence	10 (2·5)	9 (2·3)	1 (0·3)	..	..	10 (2·5)	10 (2·5)	..	..	..
Cataract	10 (2·5)	9 (2·3)	1 (0·3)	..	..	9 (2·3)	7 (1·8)	2 (0·5)	..	..
Osteonecrosis of jaw	10 (2·5)	8 (2·0)	2 (0·5)	..	..	6 (1·5)	5 (1·3)	1 (0·3)	..	..
Deep vein thrombosis	10 (2·5)	8 (2·0)	2 (0·5)	..	..	3 (0·8)	3 (0·8)	..	..	..
Spinal pain	9 (2·3)	8 (2·0)	1 (0·3)	..	..	9 (2·3)	9 (2·3)	..	..	..
Hyponatraemia	9 (2·3)	8 (2·0)	..	1 (0·3)	..	6 (1·5)	3 (0·8)	1 (0·3)	2 (0·5)	..
Rib fracture	8 (2·0)	6 (1·5)	2 (0·5)	..	..	12 (3·0)	12 (3·0)	..	..	..
Syncope	8 (2·0)	4 (1·0)	4 (1·0)	..	..	10 (2·5)	5 (1·3)	5 (1·3)	..	..
Chest pain	8 (2·0)	8 (2·0)	..	..	..	8 (2·0)	7 (1·8)	1 (0·3)	..	..
Influenza like illness	8 (2·0)	7 (1·8)	1 (0·3)	..	..	6 (1·5)	6 (1·5)	..	..	..
Urinary retention	8 (2·0)	6 (1·5)	2 (0·5)	..	..	5 (1·3)	4 (1·0)	1 (0·3)	..	..
Hypoalbuminaemia	8 (2·0)	7 (1·8)	1 (0·3)	..	..	5 (1·3)	5 (1·3)	..	..	..
COVID-19 pneumonia	8 (2·0)	2 (0·5)	2 (0·5)	..	4 (1·0)	3 (0·8)	..	3 (0·8)	..	..
Diabetes mellitus	7 (1·8)	6 (1·5)	1 (0·3)	..	..	14 (3·5)	11 (2·8)	3 (0·8)	..	..
Pain	7 (1·8)	7 (1·8)	..	..	..	7 (1·8)	6 (1·5)	1 (0·3)	..	..
Haemorrhoids	7 (1·8)	6 (1·5)	1 (0·3)	..	..	4 (1·0)	4 (1·0)	..	..	..
Cellulitis	7 (1·8)	6 (1·5)	1 (0·3)	..	..	1 (0·3)	..	1 (0·3)	..	..
Dental Caries	7 (1·8)	5 (1·3)	2 (0·5)	..	..	1 (0·3)	1 (0·3)	..	..	..
Hepatic function abnormal	6 (1·5)	4 (1·0)	1 (0·3)	1 (0·3)	..	6 (1·5)	4 (1·0)	2 (0·5)	..	..
Lower respiratory tract infection	6 (1·5)	3 (0·8)	2 (0·5)	..	1 (0·3)	6 (1·5)	6 (1·5)	..	..	..
Osteoporosis	6 (1·5)	5 (1·3)	1 (0·3)	..	..	5 (1·3)	5 (1·3)	..	..	..
Dysphagia	6 (1·5)	5 (1·3)	1 (0·3)	..	..	..	..	..	..	..
Pelvic pain	5 (1·3)	5 (1·3)	..	..	..	7 (1·8)	6 (1·5)	1 (0·3)	..	..
Blood triglycerides increased	5 (1·3)	4 (1·0)	1 (0·3)	..	..	4 (1·0)	3 (0·8)	1 (0·3)	..	..
Sepsis	5 (1·3)	1 (0·3)	2 (0·5)	..	..	3 (0·8)	..	..	2 (0·5)	1 (0·3)
Febrile neutropenia	5 (1·3)	..	2 (0·5)	3 (0·8)	..	3 (0·8)	3 (0·8)	..	..	..
Urosepsis	5 (1·3)	1 (0·3)	3 (0·8)	1 (0·3)	..	2 (0·5)	..	1 (0·3)	1 (0·3)	..

Squamous cell carcinoma of skin	5 (1·3)	4 (1·0)	1 (0·3)	..	..	2 (0·5)	..	2 (0·5)	..	..
Suspected COVID-19	5 (1·3)	3 (0·8)	1 (0·3)	..	1 (0·3)	1 (0·3)	..	1 (0·3)	..	..
Rectal haemorrhage	4 (1·0)	3 (0·8)	1 (0·3)	..	..	6 (1·5)	6 (1·5)	..	..	..
Tooth infection	4 (1·0)	4 (1·0)	..	..	..	4 (1·0)	3 (0·8)	1 (0·3)	..	..
Acute myocardial infarction	4 (1·0)	..	4 (1·0)	..	..	3 (0·8)	..	..	3 (0·8)	..
Procedural pain	4 (1·0)	3 (0·8)	1 (0·3)	..	..	3 (0·8)	3 (0·8)	..	..	..
Lipase increased	4 (1·0)	3 (0·8)	1 (0·3)	..	..	2 (0·5)	2 (0·5)	..	..	..
Hyperamylasaemia	4 (1·0)	2 (0·5)	2 (0·5)	..	..	1 (0·3)	1 (0·3)	..	..	..
<i>Pneumocystis jirovecii</i> pneumonia	4 (1·0)	1 (0·3)	3 (0·8)	..	..	..	..	..	..	..
Tibia fracture	4 (1·0)	3 (0·8)	1 (0·3)	..	..	..	..	..	..	..
Intervertebral disc protrusion	4 (1·0)	3 (0·8)	1 (0·3)	..	..	..	..	..	..	..
Cystitis	3 (0·8)	3 (0·8)	..	..	..	..	..	1 (0·3)	..	..
Leukocytosis	3 (0·8)	3 (0·8)	..	..	..	..	..	2 (0·5)	..	..
Gamma-glutamyltransferase increased	3 (0·8)	1 (0·3)	2 (0·5)	..	..	..	..	2 (0·5)	..	..
Osteoarthritis	3 (0·8)	2 (0·5)	1 (0·3)	..	..	3 (0·8)	3 (0·8)	..	..	..
Urinary tract pain	3 (0·8)	2 (0·5)	1 (0·3)	..	..	3 (0·8)	3 (0·8)	..	..	..
Angina pectoris	3 (0·8)	2 (0·5)	1 (0·3)	..	..	3 (0·8)	3 (0·8)	..	..	..
Confusional state	3 (0·8)	2 (0·5)	1 (0·3)	..	..	2 (0·5)	2 (0·5)	..	..	..
Cardiac failure	3 (0·8)	1 (0·3)	2 (0·5)	..	..	2 (0·5)	2 (0·5)	..	..	..
Pneumonitis	3 (0·8)	1 (0·3)	1 (0·3)	1 (0·3)	..	2 (0·5)	2 (0·5)	..	..	..
Decubitus ulcer	3 (0·8)	2 (0·5)	1 (0·3)	..	..	2 (0·5)	2 (0·5)	..	..	..
Ventricular extrasystoles	3 (0·8)	2 (0·5)	1 (0·3)	..	..	1 (0·3)	1 (0·3)	..	..	..
Amnesia	3 (0·8)	3 (0·8)	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Post procedural complication	3 (0·8)	2 (0·5)	1 (0·3)	..	..	1 (0·3)	1 (0·3)	..	..	..
Head injury	3 (0·8)	2 (0·5)	1 (0·3)	..	..	1 (0·3)	1 (0·3)	..	..	..
Tumour pain	3 (0·8)	3 (0·8)	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Colon cancer	3 (0·8)	1 (0·3)	2 (0·5)	..	..	1 (0·3)	..	1 (0·3)	..	..
Peripheral arterial occlusive disease	3 (0·8)	1 (0·3)	2 (0·5)	..	..	..	..	..	..	..
Polyneuropathy	3 (0·8)	2 (0·5)	1 (0·3)	..	..	..	..	..	..	..
Large intestine polyp	2 (0·5)	2 (0·5)	..	..	..	3 (0·8)	2 (0·5)	1 (0·3)	..	..
Adrenal insufficiency	2 (0·5)	2 (0·5)	..	..	..	3 (0·8)	2 (0·5)	1 (0·3)	..	..
General physical health deterioration	2 (0·5)	1 (0·3)	..	..	1 (0·3)	2 (0·5)	1 (0·3)	1 (0·3)	..	..
Ureterolithiasis	2 (0·5)	..	2 (0·5)	..	..	2 (0·5)	2 (0·5)	..	..	..
Colitis	2 (0·5)	..	2 (0·5)	..	..	2 (0·5)	2 (0·5)	..	..	..
Chronic obstructive pulmonary disease	2 (0·5)	1 (0·3)	1 (0·3)	..	..	2 (0·5)	2 (0·5)	..	..	..
Chronic kidney disease	2 (0·5)	1 (0·3)	1 (0·3)	..	..	2 (0·5)	1 (0·3)	1 (0·3)	..	..
Drug eruption	2 (0·5)	2 (0·5)	..	..	..	2 (0·5)	1 (0·3)	1 (0·3)	..	..
Pneumonia bacterial	2 (0·5)	1 (0·3)	..	..	1 (0·3)	2 (0·5)	1 (0·3)	1 (0·3)	..	..
Infection	2 (0·5)	1 (0·3)	1 (0·3)	..	..	2 (0·5)	1 (0·3)	..	..	1 (0·3)
Anaemia postoperative	2 (0·5)	1 (0·3)	1 (0·3)	..	..	1 (0·3)	1 (0·3)	..	..	..
Haematochezia	2 (0·5)	1 (0·3)	1 (0·3)	..	..	1 (0·3)	1 (0·3)	..	..	..
Myocardial ischaemia	2 (0·5)	1 (0·3)	..	..	1 (0·3)	1 (0·3)	1 (0·3)	..	..	..

Cerebral ischaemia	2 (0·5)	1 (0·3)	1 (0·3)	..	..	1 (0·3)	1 (0·3)	..	..	..
Hypophagia	2 (0·5)	1 (0·3)	1 (0·3)	..	..	1 (0·3)	..	1 (0·3)	..	..
Pneumonia aspiration	2 (0·5)	1 (0·3)	1 (0·3)	..	..	1 (0·3)	..	..	..	1 (0·3)
Skin infection	2 (0·5)	1 (0·3)	1 (0·3)	..	..	1 (0·3)	1 (0·3)	..	..	..
Femoral neck fracture	2 (0·5)	1 (0·3)	1 (0·3)	..	..	1 (0·3)	..	1 (0·3)	..	..
Diabetes mellitus inadequate control	2 (0·5)	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..
Craniocerebral injury	2 (0·5)	1 (0·3)	..	..	1 (0·3)	..	..	..	..	..
Nephrolithiasis	2 (0·5)	1 (0·3)	1 (0·3)	..	..	..	..	..	..	..
Cholecystitis	2 (0·5)	..	2 (0·5)	..	..	..	..	..	..	..
Duodenal ulcer	2 (0·5)	..	1 (0·3)	..	1 (0·3)	..	..	..	..	..
Anal incontinence	2 (0·5)	1 (0·3)	1 (0·3)	..	..	..	..	..	..	..
Left ventricular failure	2 (0·5)	1 (0·3)	1 (0·3)	..	..	..	..	..	..	..
Squamous cell carcinoma	2 (0·5)	1 (0·3)	1 (0·3)	..	..	..	..	..	..	..
Rectal cancer	2 (0·5)	1 (0·3)	1 (0·3)	..	..	..	..	..	..	..
Myelodysplastic syndrome	2 (0·5)	..	..	2 (0·5)	..	..	..	..	..	..
Bladder cancer	2 (0·5)	..	2 (0·5)	..	..	..	..	..	..	..
Acute sinusitis	1 (0·3)	1 (0·3)	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Anal abscess	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Hepatitis B reactivation	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Rash pustular	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Gastric adenoma	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Lung adenocarcinoma	1 (0·3)	..	1 (0·3)	..	..	1 (0·3)	..	1 (0·3)	..	..
Malignant melanoma	1 (0·3)	..	..	..	1 (0·3)	..	..	..	..	..
Neuroendocrine carcinoma of the skin	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Plasma cell myeloma	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Small intestine adenocarcinoma	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Squamous cell carcinoma of the tongue	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Transitional cell carcinoma	1 (0·3)	1 (0·3)	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Myelosuppression	1 (0·3)	..	..	1 (0·3)	..	..	..	..	..	..
Nightmare	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Aphasia	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Cerebrovascular accident	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Embolic stroke	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Epilepsy	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Hemiplegia	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Hypoglycaemic unconsciousness	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Ischaemic stroke	1 (0·3)	..	1 (0·3)	..	..	4 (1·0)	..	1 (0·3)	2 (0·5)	1 (0·3)
Loss of consciousness	1 (0·3)	..	..	1 (0·3)	..	4 (1.)	2 (0·5)	2 (0·5)	..	..
Cataract nuclear	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Atrial flutter	1 (0·3)	..	1 (0·3)	..	..	1 (0·3)	1 (0·3)	..	..	..
Cardiac failure acute	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Mitral valve disease	1 (0·3)	..	..	..	1 (0·3)	..	..	..	..	..
Myocardial infarction	1 (0·3)	..	1 (0·3)	..	..	4 (1·0)	2 (0·5)	2 (0·5)	..	..
Aortic dissection	1 (0·3)	..	..	1 (0·3)	..	..	..	..	..	..

Aortic stenosis	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Vasculitis	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Interstitial lung disease	1 (0·3)	..	1 (0·3)	..	..	1 (0·3)	..	..	..	1 (0·3)
Pneumothorax	1 (0·3)	..	1 (0·3)	..	..	1 (0·3)	1 (0·3)	..	..	..
Respiratory failure	1 (0·3)	..	..	..	1 (0·3)	..	..	..	..	..
Ileus	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Intestinal obstruction	1 (0·3)	..	1 (0·3)	..	..	1 (0·3)	..	..	..	..
Intestinal ulcer	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Bile duct stone	1 (0·3)	..	1 (0·3)	..	..	1 (0·3)	..	1 (0·3)	..	..
Cholangitis acute	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Cholecystitis acute	1 (0·3)	1 (0·3)	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Dermatitis	1 (0·3)	1 (0·3)	..	..	..	3 (0·8)	2 (0·5)	1 (0·3)	..	..
Osteoporotic fracture	1 (0·3)	..	1 (0·3)	..	..	3 (0·8)	3 (0·8)	..	..	..
Hydronephrosis	1 (0·3)	1 (0·3)	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Urinary tract obstruction	1 (0·3)	1 (0·3)	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Prostatitis	1 (0·3)	1 (0·3)	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Performance status decreased	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Sudden death	1 (0·3)	..	..	..	1 (0·3)	..	..	..	..	..
Death	1 (0·3)	..	..	..	1 (0·3)	4 (1·0)	..	..	..	4 (1·0)
Ejection fraction decreased	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Vitamin D decreased	1 (0·3)	1 (0·3)	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Extra-axial haemorrhage	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Femur fracture	1 (0·3)	1 (0·3)	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Gunshot wound	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Hand fracture	1 (0·3)	..	1 (0·3)	..	..	1 (0·3)	..	1 (0·3)	..	..
Hip fracture	1 (0·3)	1 (0·3)	..	..	..	2 (0·5)	..	2 (0·5)	..	..
Jaw fracture	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Joint dislocation	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Limb crushing injury	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Post procedural haematoma	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Post procedural haemorrhage	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Radiation pneumonitis	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Radius fracture	1 (0·3)	1 (0·3)	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Subdural haematoma	1 (0·3)	..	..	..	1 (0·3)	..	..	..	..	..
Toxicity to various agents	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Device occlusion	..	..	..	..	..	2 (0·5)	1 (0·3)	1 (0·3)	..	..
Sternal fracture	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Gilbert's syndrome	..	..	..	..	..	3 (0·8)	2 (0·5)	1 (0·3)	..	..
Urethral stenosis	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Postrenal failure	..	..	..	..	..	2 (0·5)	1 (0·3)	1 (0·3)	..	..
Ureteric obstruction	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Osteitis	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Portal vein thrombosis	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Cutaneous vasculitis	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Cholecystitis chronic	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Drug-induced liver injury	..	..	..	..	..	3 (0·8)	1 (0·3)	2 (0·5)	..	..
Small intestinal obstruction	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Umbilical hernia, obstructive	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..

Lower gastrointestinal haemorrhage	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Inguinal hernia	..	..	..	..	..	3 (0·8)	2 (0·5)	1 (0·3)	..	..
Gastrointestinal angiodysplasia	1 (0·3)	..	1 (0·3)	..	..	..	..	..	..	..
Colitis ulcerative	..	..	..	..	..	1 (0·3)	..	..	1 (0·3)	..
Nasal polyps	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Acute pulmonary oedema	..	..	..	..	..	2 (0·5)	..	..	..	2 (0·5)
Coronary artery disease	..	..	..	..	..	1 (0·3)	..	..	..	1 (0·3)
Atrioventricular block complete	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Acute coronary syndrome	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Limbal stem cell deficiency	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Thalamic infarction	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Intraventricular haemorrhage	..	..	..	..	..	1 (0·3)	..	..	..	1 (0·3)
Hemiparesis	..	..	..	..	..	2 (0·5)	1 (0·3)	1 (0·3)	..	..
Depressed level of consciousness	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Alcohol withdrawal syndrome	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Diabetic ketoacidosis	..	..	..	..	..	2 (0·5)	..	2 (0·5)	..	..
Anaphylactic shock	..	..	..	..	..	1 (0·3)	..	-	1 (0·3)	..
Penile squamous cell carcinoma	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Non-small cell lung cancer	..	..	..	..	..	2 (0·5)	1 (0·3)	1 (0·3)	..	..
Gastric cancer	..	..	..	..	..	2 (0·5)	1 (0·3)	1 (0·3)	..	..
Intraductal papillary mucinous neoplasm	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Colorectal cancer	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Diffuse large B-cell lymphoma	..	..	..	..	..	1 (0·3)	..	..	..	1 (0·3)
Urinary tract infection	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
pseudomonal	..	..	..	..	..	1 (0·3)	..	..	1 (0·3)	..
Septic shock	..	..	..	..	..	1 (0·3)	..	..	..	1 (0·3)
Staphylococcal sepsis	..	..	..	..	..	1 (0·3)	..	..	..	1 (0·3)
Pyelonephritis	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Osteomyelitis	..	..	..	..	..	3 (0·8)	1 (0·3)	2 (0·5)	..	..
Perirectal abscess	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Pneumococcal sepsis	..	..	..	..	..	1 (0·3)	..	..	..	1 (0·3)
Lung abscess	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Oral infection	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..
Gastroenteritis viral	..	..	..	..	..	2 (0·5)	1 (0·3)	1 (0·3)	..	..
Dacryocanaliculitis	..	..	..	..	..	1 (0·3)	..	1 (0·3)	..	..

Patients were counted once for each type of adverse event. Adverse events with an onset date, or worsening, on or after the date of first dose and up to and including 30 days following discontinuation of randomised treatment, are included. [†]Anaemia category includes anaemia, decreased haemoglobin level, decreased red-cell count, decreased haematocrit level, erythropenia, macrocytic anaemia, normochromic anaemia, normochromic normocytic anaemia, and normocytic anaemia. [‡]Neutropenia category includes febrile neutropenia, granulocyte count decreased, neutropenia, neutropenic infection, neutropenic sepsis, neutrophil count decreased, idiopathic neutropenia, and agranulocytosis. [§]Thrombocytopenia category includes platelet count decreased, platelet production decreased, thrombocytopenia, and plateletcrit decreased. [¶]Leukopenia category includes leukopenia and white blood cell count decreased.

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